

Лабораторная работа №4: Продвинутое использование git

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1. Цель работы

Получение навыков правильной работы с репозиториями git: освоение Gitflow, Conventional Commits, семантического версионирования.

2. Задание

1. Установить git-flow, Node.js, pnpm.
2. Установить commitizen et standard-changelog.
3. Создать репозиторий git-extended.
4. Настроить Conventional Commits.
5. Инициализировать git-flow.
6. Créer les releases 1.0.0 et 1.2.3.

3. Выполнение лабораторной работы

3.1. Installation

3.1.1. git-flow

```
lissombayem@fedora:/home/lissombayem/lab/lab04 x lissombayem@fedora:/home/lissombayem
[1/2] https://download.copr.fedorain ??? [ <=> ] | 0.0 B/s | 0.0 B | 00m00s
[1/2] https://download.copr.fedorain ??? [ <=> ] | 0.0 B/s | 0.0 B | 00m00s
[1/2] https://download.copr.fedorain ??? [ <=> ] | 0.0 B/s | 0.0 B | 00m01s
[1/2] https://download.copr.fedorain ??? [ <=> ] | 0.0 B/s | 0.0 B | 00m02s
[1/2] https://download.copr.fedorain ??? [ <=> ] | 0.0 B/s | 0.0 B | 00m03s
[1/2] https://download.copr.fedorain ??? [ <=> ] | 0.0 B/s | 0.0 B | 00m04s
[1/2] https://download.copr.fedorain ??? [ <=> ] | 0.0 B/s | 0.0 B | 00m05s
[1/2] https://download.copr.fedorain ??? [ <=> ] | 0.0 B/s | 0.0 B | 00m06s
[1/2] https://download.copr.fedorain ??? [ <=> ] | 0.0 B/s | 0.0 B | 00m07s
[1/2] https://download.copr.fedorain ??? [ <=> ] | 0.0 B/s | 0.0 B | 00m08s
[1/2] https://download.copr.fedorain ??? [ <=> ] | 0.0 B/s | 0.0 B | 00m09s
[1/2] https://download.copr.fedorain ??? [ <=> ] | 0.0 B/s | 0.0 B | 00m10s
[1/2] https://download.copr.fedorain ??? [ <=> ] | 0.0 B/s | 0.0 B | 00m12s
[1/2] https://download.copr.fedorain ??? [ <=> ] | 0.0 B/s | 0.0 B | 00m13s
[1/2] https://download.copr.fedorain ??? [ <=> ] | 0.0 B/s | 0.0 B | 00m14s
[1/2] https://download.copr.fedorain ??? [ <=> ] | 0.0 B/s | 0.0 B | 00m15s
[1/2] https://download.copr.fedorain ??? [ <=> ] | 0.0 B/s | 0.0 B | 00m15s
[1/2] https://download.copr.fedorain ??? [ <=> ] | 0.0 B/s | 0.0 B | 00m15s
[1/2] https://download.copr.fedorain 100% [=====] | 1.3 KiB/s | 998.0 B | 00m00s
[1/2] https://download.copr.fedorainfracloud.org/results/ 100% | 64.0 B/s | 998.0 B | 00m15s
-----
[2/2] Total 100% | 55.7 KiB/s | 57.4 KiB | 00m01s
Importation de la clé OpenPGP 0x80F63AA3 :
UserID : "elegos_gitflow (None) <elegos#gitflow@copr.fedorahosted.org>"
Empreinte : 935799B049C5C36CF39A22823CA8E680F63AA3
De : https://download.copr.fedorainfracloud.org/results/elegos/gitflow/pubkey.gpg
Is this ok [y/N]: y
La clé a été importée avec succès.
[1/3] Vérifier les fichiers des paquets 100% | 23.0 B/s | 1.0 B | 00m00s
[2/3] Préparer la transaction 100% | 2.0 B/s | 1.0 B | 00m00s
[3/3] Installation de gitflow-0:1.12.3-1.fc34.x86_64 100% | 217.9 KiB/s | 271.0 KiB | 00m01s
Terminé !
lissombayem@fedora:~/lab/lab04$
```

Установка git-flow

3.1.2. Node.js

```
lissombayem@fedora:~/lab/lab04$ sudo dnf install nodejs
Mise à jour et chargement des dépôts :
Dépôts chargés.
Le paquet "nodejs-1:22.22.0-2.fc42.x86_64" est déjà installé.

Aucune tâche à effectuer.
lissombayem@fedora:~/lab/lab04$ sudo dnf install npm
Mise à jour et chargement des dépôts :
Dépôts chargés.
Le paquet "npm-10.9.0-1.fc42.noarch" est déjà installé.

Aucune tâche à effectuer.
lissombayem@fedora:~/lab/lab04$
```

Установка Node.js

3.1.3. commitizen

```
lissombayem@fedora:/home/lissombayem/lab/lab04
lissombayem@fedora:/home/lissombayem/lab/lab04 x lissombayem@fedora:/home/lissombayem

WARN 2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +8 -11
-----
Progress: resolved 178, reused 170, downloaded 8, added 8, done

/home/lissombayem/.local/share/pnpm/global/5:
- commitizen 4.3.1
+ commitizen 4.3.2

ERR_PNPM_META_FETCH_FAIL GET https://registry.npmjs.org/pnpm: request to https://registry.npmjs.org/pnpm failed, reason
: read ECONNRESET

Update available! 10.9.0 → 10.30.3.
Changelog: https://pnpm.io/v/10.30.3
To update, run: pnpm self-update

Done in 11s using pnpm v10.9.0
WARN GET https://registry.npmjs.org/semver error (ECONNRESET). Will retry in 10 seconds. 2 retries left.
WARN GET https://registry.npmjs.org/@simple-lib%2Fchild-process-utils error (ECONNRESET). Will retry in 10 seconds. 2
retries left.
WARN 2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +15 -27
-----
Progress: resolved 166, reused 151, downloaded 15, added 15, done

/home/lissombayem/.local/share/pnpm/global/5:
- standard-changelog 7.0.1
+ standard-changelog 8.1.0

Done in 35.3s using pnpm v10.9.0
lissombayem@fedora:~/lab/lab04$
```

Установка commitizen

3.2. Création du dépôt

```
lissombayem@fedora:~/lab/lab04$ gh repo create git-extended --public
GraphQL: Name already exists on this account (createRepository)
lissombayem@fedora:~/lab/lab04$ 
lissombayem@fedora:~/lab/lab04$ git clone git@github.com:odilonnathanael/git-extended.git
cd git-extended
echo "# git-extended" > README.md
git add README.md
git commit -m "first commit"
git push -u origin master
Clonage dans 'git-extended'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (4/4), done.
remote: Total 6 (delta 0), reused 6 (delta 0), pack-reused 0 (from 0)
Réception d'objets: 100% (6/6), fait.
Sur la branche master
Votre branche est à jour avec 'origin/master'.

rien à valider, la copie de travail est propre
la branche 'master' est paramétrée pour suivre 'origin/master'.
Everything up-to-date
lissombayem@fedora:~/lab/lab04/git-extended$
```

3.3. Configuration package.json

```
lissombayem@fedora:~/lab/lab04/git-extended$ cat package.json
{
  "name": "git-extended",
  "version": "1.0.0",
  "description": "Git repo for educational purposes",
  "main": "index.js",
  "repository": "git@github.com:odilonnathanael/git-extended.git",
  "author": "Odilon Nathanael <odilonnathanael@yahoo.com>",
  "license": "CC-BY-4.0",
  "config": {
    "commitizen": {
      "path": "cz-conventional-changelog"
    }
  }
}
```

package.json

3.4. Commit avec commitizen

```
lissombayem@fedora:/home/lissombayem/lab/lab04/git-extended x lissombayem@fedora:/home/lissombayem
create mode 100644 node_modules/.pnpm/wcwidth@1.0.1/node_modules/wcwidth/.npmignore
create mode 100644 node_modules/.pnpm/wcwidth@1.0.1/node_modules/wcwidth/LICENSE
create mode 100644 node_modules/.pnpm/wcwidth@1.0.1/node_modules/wcwidth/Readme.md
create mode 100644 node_modules/.pnpm/wcwidth@1.0.1/node_modules/wcwidth/combining.js
create mode 100644 node_modules/.pnpm/wcwidth@1.0.1/node_modules/wcwidth/docs/index.md
create mode 100644 node_modules/.pnpm/wcwidth@1.0.1/node_modules/wcwidth/index.js
create mode 100644 node_modules/.pnpm/wcwidth@1.0.1/node_modules/wcwidth/package.json
create mode 100644 node_modules/.pnpm/wcwidth@1.0.1/node_modules/wcwidth/test/index.js
create mode 120000 node_modules/.pnpm/which@1.3.1/node_modules/isexe
create mode 100644 node_modules/.pnpm/which@1.3.1/node_modules/which/CHANGELOG.md
create mode 100644 node_modules/.pnpm/which@1.3.1/node_modules/which/LICENSE
create mode 100644 node_modules/.pnpm/which@1.3.1/node_modules/which/README.md
create mode 100755 node_modules/.pnpm/which@1.3.1/node_modules/which/bin/which
create mode 100644 node_modules/.pnpm/which@1.3.1/node_modules/which/package.json
create mode 100644 node_modules/.pnpm/which@1.3.1/node_modules/which/which.js
create mode 100644 node_modules/.pnpm/word-wrap@1.2.5/node_modules/word-wrap/LICENSE
create mode 100644 node_modules/.pnpm/word-wrap@1.2.5/node_modules/word-wrap/README.md
create mode 100644 node_modules/.pnpm/word-wrap@1.2.5/node_modules/word-wrap/index.d.ts
create mode 100644 node_modules/.pnpm/word-wrap@1.2.5/node_modules/word-wrap/index.js
create mode 100644 node_modules/.pnpm/word-wrap@1.2.5/node_modules/word-wrap/package.json
create mode 120000 node_modules/.pnpm/wrap-ansi@6.2.0/node_modules/ansi-styles
create mode 120000 node_modules/.pnpm/wrap-ansi@6.2.0/node_modules/string-width
create mode 120000 node_modules/.pnpm/wrap-ansi@6.2.0/node_modules/strip-ansi
create mode 100755 node_modules/.pnpm/wrap-ansi@6.2.0/node_modules/wrap-ansi/index.js
create mode 100644 node_modules/.pnpm/wrap-ansi@6.2.0/node_modules/wrap-ansi/license
create mode 100644 node_modules/.pnpm/wrap-ansi@6.2.0/node_modules/wrap-ansi/package.json
create mode 100644 node_modules/.pnpm/wrap-ansi@6.2.0/node_modules/wrap-ansi/readme.md
create mode 100644 node_modules/.pnpm/wrappy@1.0.2/node_modules/wrappy/LICENSE
create mode 100644 node_modules/.pnpm/wrappy@1.0.2/node_modules/wrappy/README.md
create mode 100644 node_modules/.pnpm/wrappy@1.0.2/node_modules/wrappy/package.json
create mode 100644 node_modules/.pnpm/wrappy@1.0.2/node_modules/wrappy/wrappy.js
create mode 120000 node_modules/cz-conventional-changelog
create mode 100644 pnpm-lock.yaml
lissombayem@fedora:~/lab/lab04/git-extended$
```

git cz

3.5. Initialisation git-flow

```
lissombayem@fedora:~$ sudo -i
[sudo] Mot de passe de lissombayem :
root@fedora:~# dnf install git
Mise à jour et chargement des dépôts :
Dépôts chargés.
Le paquet "git-2.53.0-1.fc42.x86_64" est déjà installé.

Aucune tâche à effectuer.
root@fedora:~#
```

git flow init

3.6. Release 1.0.0

```
remote:
To github.com:odilonnathanael/git-extended.git
* [new branch]      feature/feature_branch -> feature/feature_branch
lissombayem@fedora:~/lab/lab04/git-extended$ git push --tags
Everything up-to-date
lissombayem@fedora:~/lab/lab04/git-extended$ standard-changelog
git add package.json CHANGELOG.md
git commit -am 'chore(site): update changelog'
git flow release finish 1.2.3
git push --all
git push --tags
gh release create v1.2.3 -F CHANGELOG.md
[feature/feature_branch 24d7a11] chore(site): update changelog
 1 file changed, 7 insertions(+)
 create mode 100644 CHANGELOG.md
egrep: warning: egrep is obsolescent; using grep -E
Branch 'V1.2.3' does not exist and is required.
Énumération des objets: 4, fait.
Décompte des objets: 100% (4/4), fait.
Compression par delta en utilisant jusqu'à 3 fils d'exécution
Compression des objets: 100% (3/3), fait.
Écriture des objets: 100% (3/3), 1.17 KiO | 401.00 KiO/s, fait.
Total 3 (delta 1), réutilisés 0 (delta 0), réutilisés du paquet 0 (depuis 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:odilonnathanael/git-extended.git
 d21a190..24d7a11 feature/feature_branch -> feature/feature_branch
Everything up-to-date
https://github.com/odilonnathanael/git-extended/releases/tag/v1.2.3
lissombayem@fedora:~/lab/lab04/git-extended$
```

Release 1.0.0

3.7. Feature

```
lissombayem@fedora:~/lab/lab04/git-extended$ git flow feature start feature_branch
echo "New feature" > feature.txt
git add feature.txt
git cz
egrep: warning: egrep is obsolescent; using grep -E
Basculement sur la nouvelle branche 'feature/feature_branch'

Summary of actions:
- A new branch 'feature/feature_branch' was created, based on 'develop'
- You are now on branch 'feature/feature_branch'

Now, start committing on your feature. When done, use:

    git flow feature finish feature_branch

cz-cli@4.3.2, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: feat:      A new feature
? What is the scope of this change (e.g. component or file name): (press enter to skip) feature
? Write a short, imperative tense description of the change (max 85 chars):
  (15) add new feature
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[feature/feature_branch 941009f] feat(feature): add new feature
 1 file changed, 1 insertion(+)
 create mode 100644 feature.txt
lissombayem@fedora:~/lab/lab04/git-extended$
```

Feature finish

3.8. Release 1.2.3

```
lissombayem@fedora:~/lab/lab04/git-extended$ git flow release start 1.0.0
standard-changelog --first-release
git add CHANGELOG.md
git commit -am 'chore(site): add changelog'
git flow release finish 1.0.0
egrep: warning: egrep is obsolescent; using grep -E
Basculement sur la nouvelle branche 'V1.0.0'

Summary of actions:
- A new branch 'V1.0.0' was created, based on 'develop'
- You are now on branch 'V1.0.0'

Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

    git flow release finish '1.0.0'

[V1.0.0 5508da1] chore(site): add changelog
 1 file changed, 6 insertions(+)
 create mode 100644 CHANGELOG.md
egrep: warning: egrep is obsolescent; using grep -E
Basculement sur la branche 'master'
Votre branche est à jour avec 'origin/master'.
^XMerge made by the 'ort' strategy.
CHANGELOG.md | 6 +++++
 1 file changed, 6 insertions(+)
 create mode 100644 CHANGELOG.md
fatal : pas de message pour l'étiquette ?
Tagging failed. Please run finish again to retry.
lissombayem@fedora:~/lab/lab04/git-extended$
```

Release 1.2.3

4. Выводы

В ходе работы были освоены Gitflow, Conventional Commits et SemVer.

5. Список литературы

1. Документация git-flow.
2. Спецификация Conventional Commits.